Lesson: Who Designs a Product—and Who Owns Its Story?

Core question for students: Is design the shape of an object, the system that makes it usable, the patent that protects it, or the economic machine that distributes it? This session uses Edison-era products to examine creativity, appropriation, and the hidden social infrastructure behind familiar objects.

Learning objectives

By the end, students should be able to:

- Distinguish **invention**, **innovation**, **design**, **appropriation**, and **commercialization**.
- Analyze products as assemblages of form, function, manufacture, law, capital, and cultural meaning.
- Critically assess how "authorship" is constructed around technologies.
- Apply these questions to their own design practice: interior, industrial, graphic, product, or speculative design.

Framing: design beyond the object

For designers, a product is never only an artifact. A light bulb is glass, tungsten, socket geometry, electrical infrastructure, pricing, labor, energy policy, and a cultural metaphor for "ideas." A projector is optics, film format, exhibition space, ticket sales, censorship, and collective spectatorship.

This lesson therefore treats design at three levels:

Level	Question	Example
Artifact	What is it made of, and how does it work?	Filament, glass envelope, screw base
System	What infrastructure makes it function?	Generators, wiring, meters, utilities
Political economy	Who profits, controls access, and defines authorship?	Patent holders, banks, utilities, distributors

Case 1: The incandescent lamp

The common claim that Edison "invented the light bulb" is historically incomplete. Edison's laboratory developed a commercially practical incandescent lighting system, but earlier inventors had worked on incandescent lamps, and the U.S. Supreme Court later invalidated a

patent belonging to William Sawyer and Albon Man; the Court did not decide that Edison invented the light bulb. [1]

From a design perspective, Edison's major contribution was **system design**: a durable lamp, generators, distribution wiring, meters, sockets, and a business model that could make electric lighting viable. This is crucial for product and industrial designers: the breakthrough was not only the object, but the interoperable ecosystem around it.

Discussion prompt: If you redesign a lamp today, are you designing a light source, a domestic ritual, an energy interface, or a data-connected service?

Case 2: The phonograph

Edison's phonograph is a stronger case of genuine invention, but even here authorship was collective. Edison sketched the concept after work on telephone technology; machinist John Kruesi built the first tinfoil machine from Edison's drawings. Edison filed on December 24, 1877, and received the patent on February 19, 1878. [2] [3]

French inventor Charles Cros had earlier proposed a similar recording principle in April 1877, though he did not build a working device. Léon Scott's 1857 phonautograph also anticipated aspects of sound recording. [4] [2]

Design lesson: The phonograph shows how a concept may circulate before one actor turns it into a working, manufacturable product. Creativity is often a relay, not a single act.

Studio exercise: Redesign the phonograph for a contemporary social need—memory preservation, oral history, accessibility, or community archiving—while identifying who would own the data, hardware, and distribution.

Case 3: The Kinetoscope and cinema projector

The motion-picture case is especially useful because it separates invention from the eventual medium. Edison commissioned William Kennedy Laurie Dickson to develop a motion-picture apparatus; Dickson performed much of the experimentation. The result was the Kinetograph camera and Kinetoscope, a coin-operated peephole device for one viewer at a time. [5] [6] [7] [8]

The public cinema projector was not Edison's original design goal. After the Lumière brothers demonstrated their Cinématographe in Paris in 1895—a device that could film, print, and project—Edison's organization added projection to reach larger audiences. [9] [10] [5]

This is a powerful design contrast:

- **Kinetoscope:** private, boxed, individual, coin-operated.
- **Cinématographe:** public, projected, social, portable.

The difference is not merely technical. It changes the social form of the experience: from a solitary curiosity to collective cinema, from a machine in an arcade to an institution that shapes public culture.

Design prompt: Choose a contemporary screen technology—VR headset, smartphone, interactive installation—and redesign it by shifting its social scale: private to public, personal to communal, owned to shared.

Case 4: Radio and the myth of single authorship

Radio is essential for correcting the Edison-centered narrative. Edison was not the principal inventor of radio. Radio emerged through the work of figures including James Clerk Maxwell, Heinrich Hertz, Oliver Lodge, Nikola Tesla, Guglielmo Marconi, and others. Marconi is often publicly remembered as "the inventor of radio," while Tesla, Lodge, and many contributors were marginalized in popular accounts.

For design students, radio demonstrates that a product category can be **socially attributed** to one name even when it results from distributed scientific and engineering work. The "radio set" as a domestic object was also a design act: it translated invisible electromagnetic communication into a cabinet, dial, speaker, living-room ritual, national broadcast culture, and advertising economy.

Critical question: When a company owns the brand, patent, spectrum license, transmitter network, and retail channel, who is the real author of the user experience?

Case 5: Edison's patent system and the "Edison Trust"

The Motion Picture Patents Company, or "Edison Trust," is the clearest example of design power becoming market power. Edison and allied patent holders pooled patents over cameras, projectors, and film stock, then controlled licensing across production, distribution, and exhibition. [11] [12] [13]

The trust attempted to make independent filmmakers dependent on licensed equipment and film stock. Independent producers faced extensive litigation; the system was dissolved under U.S. antitrust law in 1915, with the court rejecting the use of patents as a weapon to disable competitors. [14]

For designers, this raises an uncomfortable point: the same standardized film format that enabled industrial efficiency also enabled control. Standardization can democratize production by making tools compatible, but it can also create chokepoints when one actor owns the standard.

Banking, capital, and the designer's blind spot

Edison's enterprises were not financed only by sales. J.P. Morgan and Anthony Drexel helped finance and consolidate Edison's electrical businesses; Edison General Electric was formed in 1889, then merged with Thomson-Houston in 1892 to create General Electric. [15] [16] [17]

The result was paradoxical: Edison lost operational control of the company that carried part of his name. Thomson-Houston management led the new General Electric, while financiers shaped the corporate structure. [16] [17]

This is the “lower-level” design discussion students need: every product exists inside a financial architecture. A chair is affected by material suppliers and retail margins; a medical device by insurance systems; a lamp by utility regulation; an app by platform fees and venture capital.

Key idea: Banking does not merely fund design; it can determine which designs survive, who receives credit, and which futures become economically visible.

Appropriation versus attribution

Use this vocabulary carefully in class:

Term	Meaning	Example
Invention	Creating a new technical principle or working device	Edison’s working tinfoil phonograph
Improvement	Refining performance, usability, or manufacturability	Edison’s practical lighting system
Appropriation	Taking credit, value, or control from others’ contributions	Corporate attribution of team and predecessor work to a single “genius”
Patent capture	Using legal rights to control a wider market than the invention itself	Broad claims and the MPPC licensing system
Standard-setting	Establishing shared technical rules	Film gauge, electrical sockets, broadcast formats

The ethical issue is not that Edison patented inventions—patents can protect genuine creative labor. The issue is when patents, branding, litigation, and finance convert one actor’s contribution into control over an entire field.

Creative techniques for the studio

1. Product biography

Trace one object from raw material to disposal. Include designers, factory workers, patent holders, investors, distributors, users, and waste streams.

2. Authorship map

Draw a network diagram: inventors, assistants, funders, manufacturers, regulators, and users. Mark where public history gives credit and where it disappears.

3. Counter-design

Redesign the product under a different ownership model: open source, cooperative, public utility, repairable commons, or community-owned infrastructure.

4. Format politics

Take a format—35 mm film, MP3, PDF, USB-C, a building typology—and ask: What does it make easy? What does it make impossible? Who benefits from its standardization?

5. Speculative patent

Write a patent-style document for an everyday social practice: “a method for sharing tools,” “a system for collective memory,” “a device for slowing attention.” Then ask what happens when it is owned by a bank, a platform, or a public institution.

Assignment

Brief: Select one product—lamp, phonograph, Kinetoscope, cinema projector, radio, telephone, or a contemporary equivalent. Produce:

1. A redesigned artifact or system.
2. An authorship and value-flow map.
3. A one-page critical analysis of ownership, access, labor, and finance.
4. A prototype or visual proposal showing an alternative economic model.

The goal is not to condemn innovation, but to make designers aware that every object carries a political economy. Design is not only making things desirable; it is deciding who may participate in their creation, use, repair, and benefit. [5] [11] [14] [15]

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